

Sai Srujan Vemula

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AI/ML Engineer | Agentic AI and RAG Systems | Production MLOps on AWS | Model Risk and Explainability

Profile Summary

AI/ML Engineer with 4+ years building production machine learning and generative AI systems across banking, fintech, and telecom. Currently own a multi-agent RAG platform on **LangGraph and AWS Bedrock AgentCore** serving **400+ internal users** at **91% retrieval precision**, with RAGAS and TruLens evaluation gates in CI/CD. Trained fraud and credit risk models on **15M+ transactions** with SHAP explainability for regulatory review. Google Professional Machine Learning Engineer and AWS Certified Machine Learning Engineer Associate.

Work Experience

AI/ML Engineer, M&T Bank, Bridgeport, CT Mar 2025 – Present

- Architected a multi-agent **RAG platform** on LangGraph and AWS Bedrock AgentCore with planner, retriever, executor, and validator agents, MCP tool-calling contracts, and hybrid BM25 and dense retrieval, raising retrieval precision from **72% to 91%** for **400+ internal users** at sub-second retrieval latency.
- Defined the bank's **agentic AI standard** with Model Risk Management and Compliance: MCP server contracts, Bedrock Guardrails enforcement, OAuth2/JWT least-privilege access, and human-in-the-loop approval for high-risk actions, adopted across internal LLM and agent surfaces.
- Instrumented **RAGAS and TruLens evaluation gates** into Jenkins CI/CD with tool-call trace debugging so every retrieval or agent change passes automated regression suites before release, cutting production guardrail failures **40%**.
- Deployed the platform on **AWS EKS** with Docker, Terraform, and ArgoCD, running agents on the Bedrock AgentCore serverless runtime and meeting a **99.9% availability SLA** under production load.
- Trained **fraud and credit risk models** on **15M+ transactions** with XGBoost and PyTorch, adding per-prediction SHAP attribution that improved detection accuracy **12%** and gave risk reviewers full decision traceability.
- Built a React and TypeScript **AI monitoring dashboard** on OpenTelemetry traces and ELK logs, surfacing agent metrics, tool-call traces, and evaluation scores so on-call engineers diagnose failures in minutes rather than hours.

Machine Learning Engineer, Charter Communications, Stamford, CT May 2024 – Feb 2025

- Built an **agentic LLM support assistant** with LangChain orchestration, Pinecone-backed RAG, and intent classification routing to ServiceNow, deflecting **28%** of inbound ticket volume.
- Developed **churn and segmentation models** on **8M+ subscriber records** with XGBoost, TensorFlow, and uplift modeling validated through A/B tests, with outputs feeding retention campaign targeting.
- Engineered streaming pipelines on **Kafka, Spark, and Prefect** that ingest network equipment telemetry at **95% reconciliation accuracy** and apply LSTM and Autoencoder anomaly models, flagging failures **24 to 72 hours** before service impact.
- Maintained **8+ production model pipelines** behind an MLflow registry with drift monitoring and automated retraining triggers.
- Developed **LLM-as-Judge evaluation harnesses** with ground-truth sets, offline benchmarks, and failure-mode analysis to gate fine-tuning rollouts before each release.
- Served Llama, Mistral, and Qwen models on **vLLM and Triton Inference Server** with KV-cache tuning, holding **sub-200ms p99** time-to-first-token under peak load.

Machine Learning Engineer, M2P Fintech, Chennai, India Feb 2022 – Dec 2023

- Built a real-time **account takeover detection** system on **2M+ daily financial events**, combining sliding-window behavioral features with supervised scoring and serving TensorRT and ONNX compiled models over gRPC at **sub-50ms p99**.
- Launched XGBoost and LightGBM **payment risk scoring** into Java and C#/.NET microservices over gRPC, reducing annual fraud losses **8%** with per-decision SHAP outputs for regulatory review.
- Automated **KYC and AML document processing** with Hugging Face Transformers and spaCy for entity extraction and classification, emitting Pydantic-validated JSON with PII encryption and cutting manual processing time **45%**.

- Packaged FastAPI **prediction microservices** in Docker on Kubernetes with MLflow model registry, drift monitoring, and automated rollback for zero-downtime releases through regulatory review cycles.
- Shipped streaming **ETL pipelines** on Spark, Airflow, and Hadoop feeding fraud-screening feature stores, with point-in-time correct reconciliation checks on every batch load.
- Standardized **REST and gRPC contracts** across payment platform services, replacing point-to-point integrations with versioned interfaces and cutting integration setup time **40%**.

Technical Skills

Languages Python, SQL, TypeScript, Java, C#/.NET, Bash

Agent AI and LLMs LangChain, LangGraph, AWS Bedrock (AgentCore, Guardrails, Knowledge Bases), MCP, OpenAI API, Anthropic Claude API, Llama, Mistral, Qwen, RAG, Hybrid BM25/Vector Retrieval, Cross-Encoder Re-Ranking, Prompt Engineering, DPO and SFT Fine-Tuning

Evaluation and Model Risk RAGAS, TruLens, LLM-as-Judge, SHAP, LIME, MLflow, Drift Monitoring, Bias Evaluation, Human-in-the-Loop Workflows, Model Governance Documentation

ML and Inference PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, spaCy, vLLM, Triton Inference Server, TensorRT, ONNX

Cloud and MLOps AWS (SageMaker, EKS, Lambda, S3, Kinesis, CloudWatch), Azure (OpenAI, Azure ML, AI Search), GCP (Vertex AI), Docker, Kubernetes, Terraform, Jenkins, ArgoCD, GitHub Actions, OpenTelemetry, ELK Stack, Prometheus, Grafana

Data and APIs Kafka, Spark, PySpark, Airflow, Prefect, Redis, PostgreSQL, Snowflake, FAISS, Pinecone, pgvector, OpenSearch, FastAPI, gRPC, REST, Pydantic, OAuth2/JWT, React

Education

Master of Science in Computer Science, Quinnipiac University, Hamden, Connecticut, USA

Thesis: Context-Aware Security Requirements Analysis Using Large Language Models. Blind expert-evaluated comparison of GPT-4o, Gemini 2.5 Flash, and DeepSeek on security abuse-case generation from SRS documents across healthcare, finance, and e-commerce; designed a keyword-based context-extraction method that let a 64K-context model match a 900K-context model on expert-scored quality.

Bachelor of Technology in Computer Science and Engineering, Lovely Professional University, Punjab, India

Certifications

Google Professional Machine Learning Engineer

Google Cloud

AWS Certified Machine Learning Engineer Associate (MLA-C01)

Amazon Web Services

Databricks Certified Machine Learning Professional

Databricks

IBM Generative AI Engineering Professional Certificate

IBM